Name:_____ Period:_____

Entry Ticket: Bowen's Reaction Series

Today we will be learning how magma forms and how its cooling rate and mineral composition shape igneous rocks so we can explain why Yosemite's granite has large crystals while Hawai'i's basalt and obsidian have tiny crystals or none at all.

_____**1. (What controls the order)** Bowen found that minerals form from magma in a predictable order. What controls this order?

A. How quickly the magma moves underground
B. The temperature of the magma as it cools
C. How much water mixes into the magma
D. The size of the crystals that form first

_____**2. (Why the minerals change)** Why does a different mineral form after olivine?

A. The magma heats back up, which melts the olivine crystals that formed.
B. The magma gains new elements from the air above it as it cools.
C. Olivine uses up iron and magnesium, so the next mineral uses leftover elements.
D. The olivine crystals slowly change their atoms and turn directly into quartz.

_____**3. (The temperature pattern)** Which statement best describes minerals that form at high temperatures?

A. They are felsic minerals, with less iron and magnesium.
B. They are mafic minerals, rich in iron and magnesium.
C. They contain the same elements, in the same amounts, as low-temperature minerals.
D. They are made only of silicon and oxygen.

4. (The end of the series) Quartz is the last mineral to form. Why is quartz the mineral you most often find in beach sand?

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